Outputs Explanation

The document goes through every output file produced by the programs in the source code archive and explains what is inside it, how to read it, and what it told us. Figures are described but not shown here; the files themselves sit in the numbered output folders.

string_prep_and_grn_overlap.py

STRING is a database of protein interactions, and it is a useful second opinion because it uses completely different evidence from the GRN. The GRN only knows about transcription factors controlling genes, whereas STRING also knows about proteins that physically bind each other. The STRING analysis itself runs on the STRING website, so the script does the two Python steps around it: preparing the gene lists that go in, and checking the results that come back against the GRN.

- **string_input_human_symbols.txt:** the full list of significant DEGs converted to human gene symbols, one per line, formatted so it can be pasted straight into the STRING website. The conversion is needed because human STRING is much better annotated than the zebrafish version.
- **string_input_UP_in_MUT_human.txt:** the same but only for the genes that go up in the mutant, 917 symbols.
- **string_input_DOWN_in_MUT_human.txt:** and only the genes that go down, 2,879 symbols. Splitting them mattered a lot. Running everything together gave a blurred result because the down genes outnumber the up genes so heavily, whereas the separate runs gave two clean and completely different stories: cell cycle and DNA replication for the down list, and tissue remodelling and stress for the up list.
- **grn_string_overlap_summary.csv:** the overlap counts between the main STRING terms and the GRN target sets. This is the file that shows the two methods are not just telling similar stories in general terms but are actually pointing at the same individual genes, for example that the genes STRING puts in its cell cycle term are the same genes the GRN lists as E2F targets.